

# TRACE Test — Should AI Make This Decision?

Scenario: AI in Loan Approval

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## 1. Scenario Description

AI in loan approval refers to the use of machine learning models by banks and NBFCs to assess creditworthiness and decide whether to approve or reject a loan application, often replacing or supplementing traditional credit-score-based underwriting. These models draw on transaction history, repayment behaviour, and sometimes alternative data such as utility payments or app usage patterns to predict default risk far faster than a human loan officer reviewing a file manually. The efficiency gain is real, but so is the risk: because the models are trained on historical lending data, they can inherit and reproduce past patterns of discrimination, such as systematically lower approval rates or higher interest rates for applicants from certain postal codes, castes, or genders, even when protected attributes are not used directly as inputs.

## 2. Applying the TRACE Test

**T — Task:** Initial credit-risk scoring, document verification, and preliminary eligibility filtering can be automated, since these involve structured, high-volume calculations that a model can perform consistently and quickly across thousands of applications.

**R — Risk:** The chief harms are financial exclusion of already disadvantaged groups, such as first-time borrowers, informal-sector workers, rural applicants, and minorities, who may be flagged as high-risk due to proxy variables correlated with historical bias rather than genuine repayment capacity.

**A — Authority:** A qualified loan officer must be able to review and override an AI-generated rejection or approval, especially for borderline cases or first-time applicants with thin credit files, and this override authority needs to be a formal, exercised part of the process rather than a theoretical option.

**C — Contestability:** Applicants who are rejected must be told that an automated system contributed to the decision and given specific, understandable reasons, along with a simple process to appeal or request manual reassessment, since a black-box rejection with no explanation leaves people with no way to correct errors.

**E — Evidence:** Evidence shows AI credit models can outperform traditional scoring on default prediction accuracy, but regulatory investigations, including cases examined by the US Consumer Financial Protection Bureau, have found some models produce disparate outcomes across demographic groups even while being statistically accurate overall, meaning better predictive power does not automatically mean fairer outcomes than a well-regulated human process.

## 3. AI Assistance vs. Human Decision

**AI should assist with:** preliminary risk scoring, document and identity verification, and flagging applications that are clear approvals or clear rejections under transparent, pre-set criteria.

**Must remain human-led:** final approval decisions for borderline applicants, any rejection that will be communicated to the applicant, and periodic fairness audits of the model's outcomes across demographic groups.

## 4. Non-Negotiable Ethical Safeguard

Every loan rejection influenced by the AI model must include a clear, applicant-facing explanation of the key factors behind the decision, and the lender must run regular disparate-impact audits comparing approval and interest-rate outcomes across demographic groups, with a human compliance officer authorised to halt or adjust the model if bias is detected.

## 5. Conclusion

AI in loan approval should be partially automated, not fully automated or fully human-led. Full automation removes accountability for decisions that materially affect people's access to housing, education, and business capital, while a fully manual process sacrifices the speed and consistency that a well-governed model can offer. The appropriate balance uses AI for preliminary scoring and document processing under mandatory human review of borderline and rejected cases, backed by explanation requirements and regular fairness audits, so lending efficiency does not come at the cost of equitable access to credit.

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## References

Consumer Financial Protection Bureau. (2023). CFPB issues guidance on credit denials by lenders using artificial intelligence. CFPB.  
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